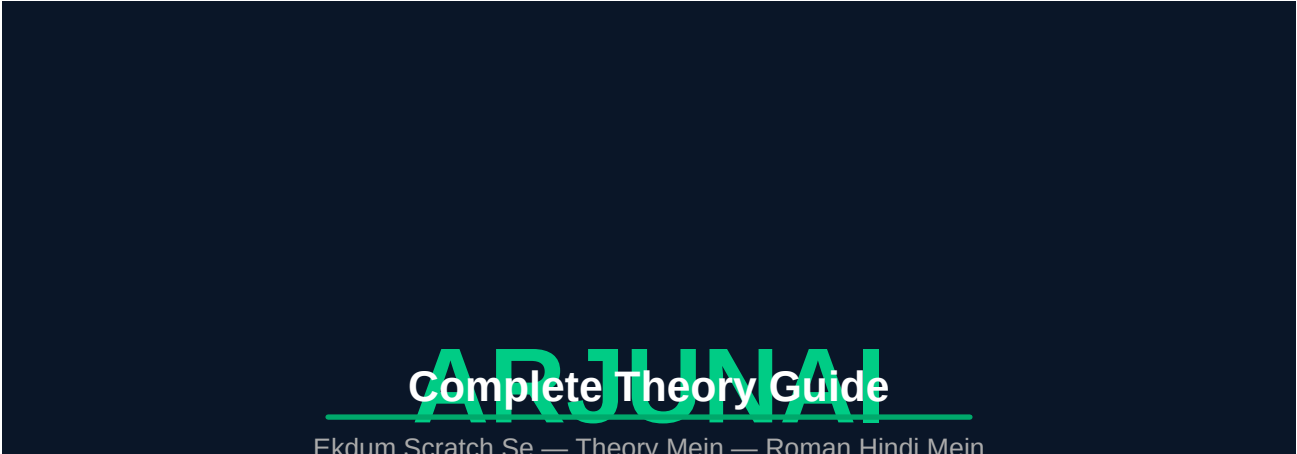




Kya Hai?	Frontend	Backend	AI Agents	System Prompt	Cache	6 Weeks	Deploy
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Finowings AI | NSE/BSE Indian Stock Market | Web Chatbot | 6 Week Plan

1. ARJUNAI KYA HAI?

Basics — Bilkul Simple Language Mein

Sabse Simple Explanation

ArjunAI ek web chatbot hai. User apne browser mein jaata hai — jaise Chrome ya Firefox — ek website kholta hai, aur wahan ek chat box hota hai. User us chat box mein stock se related sawaal type karta hai — jaise 'Reliance Industries ka P/E ratio kya hai?' ya 'Nifty 50 ka outlook kya hai?' — aur ArjunAI jawab deta hai. Bilkul WhatsApp jaisa — lekin yahan doosri taraf koi insaan nahi, ek AI hai.

Andar Se Kya Hota Hai?

Upar se dekhne mein yeh sirf ek chat box lagta hai. Lekin andar se ek poora system kaam karta hai. Teen cheezein milke ek chatbot banati hain.

Analogy

Socho ek restaurant hai. Customer dining room mein baitha hai — woh menu dekhta hai, order deta hai, khana aata hai. Yeh frontend hai. Kitchen mein sab kaam hota hai — chef khana banata hai. Yeh backend hai. Chef ke paas recipes hain — woh unke hisaab se khana banata hai. Yeh AI agents hain.

Poora System — Diagram

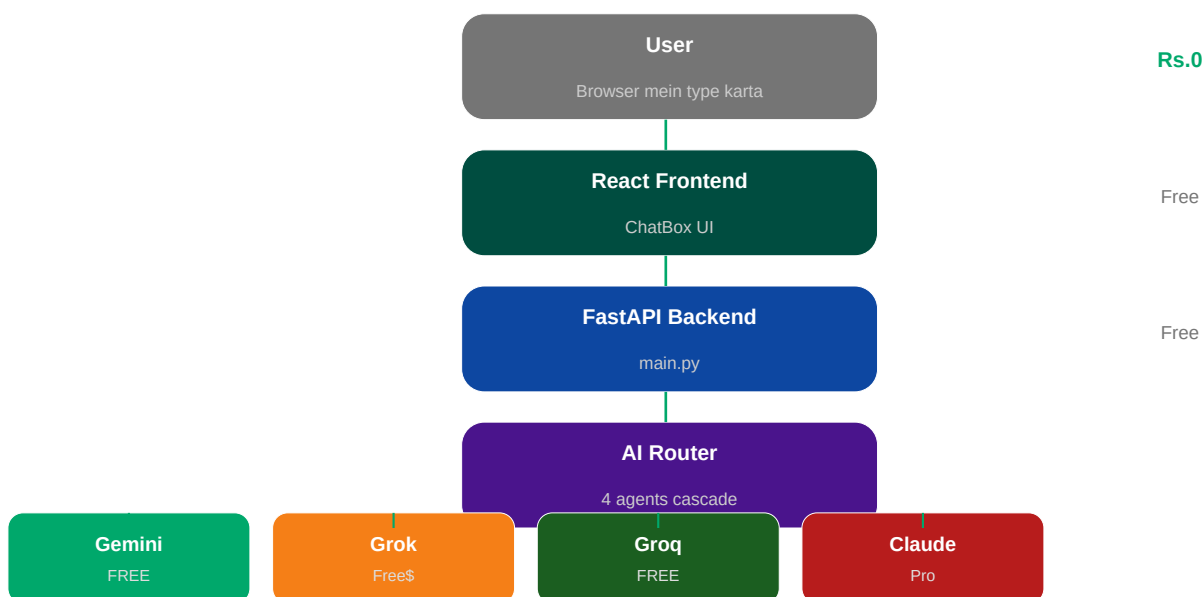


Diagram mein upar se neeche ka flow hai. User browser mein sawaal type karta hai. React Frontend us sawaal ko FastAPI Backend ke paas bhejta hai. FastAPI Backend AI Router ko call karta hai. AI Router decide karta hai kaunsa AI agent use karna hai — pehle Gemini try hota hai kyunki free hai, phir Grok, phir Groq, phir Claude. Jawab wapas aata hai React pe aur screen pe dikhta hai.

ArjunAI Ka Naam Kyun?

Investing.com ne apna AI chatbot 'WarrenAI' rakha — Warren Buffett ke naam pe. Finowings AI ka chatbot Indian investors ke liye hai — isliye Mahabharata ke sabse accurate archer Arjun ke naam pe 'ArjunAI' rakha. Arjun hamesha sahi nishana lagata tha — waise hi ArjunAI hamesha accurate financial

information deta hai.

KEY POINT ArjunAI SIRF Indian stock market ke sawaalon ka jawab deta hai. Crypto, recipes, weather, cricket — kuch bhi aur poochho toh woh politely refuse kar deta hai. Yeh intentional hai.

2. FRONTEND KYA HAI?

Jo User Screen Pe Dekhta Hai

Frontend Ki Definition

Frontend woh sab kuch hai jo user apni screen pe dekhta hai. Chat ke bubbles, input box, header mein ArjunAI ka naam, green dot jo dikhata hai ki system live hai, typing animation jab AI jawab soch raha hota hai — yeh sab frontend hai. Frontend browser mein chalta hai — user ke computer ya phone pe. Server pe nahi.

React — Kya Hai Aur Kyun Use Karte Hain?

React ek JavaScript library hai jo Facebook ne 2013 mein banaya tha aur ab open source hai. React ka main concept hai components. Component matlab ek chhota sa UI piece — jaise ek button, ek message bubble, ek header. Har component ek alag file mein hota hai. Sab components milke poora page banta hai. Iska faida yeh hai ki ek baar ChatBubble component banao, use har jagah reuse kar sakte ho bina dobara likhne ke. ArjunAI mein ChatBox ek component hai, Message ek component hai, Header ek component hai, Suggestions ek component hai.

Analogy

Socho LEGO blocks. Har block alag shape ka hota hai. Inhe combine karke koi bhi cheez bana sakte ho. React mein har component ek LEGO block hai. ChatBox + Message + Header + Input = poora ArjunAI chatbot UI.

Tailwind CSS — Kya Hai Aur Kyun?

CSS woh language hai jo batata hai ki cheezein kaise dikhni chahiye — rang kya ho, size kya ho, position kahan ho. Tailwind CSS ek special CSS framework hai jisme aap HTML mein hi directly classes likhte ho. For example, bg-gray-900 likhne se dark grey background aa jaata hai. text-green-400 likhne se green color text. rounded-xl likhne se rounded corners. Alag se CSS file likhne ki zarurat nahi — sab HTML mein hi ho jaata hai. ArjunAI ka dark theme Tailwind classes se banegi — bilkul Finowings AI ke brand ke saath.

Vite — Kya Hai Aur Kyun?

Vite ek build tool hai. Jab aap React mein code likhte ho, browser directly woh nahi samajh sakta. Vite us code ko process karke browser-friendly format mein convert karta hai. Development ke time Vite ka ek aur faida hai — hot reload. Aap code save karo, aur browser mein turant result dikhai deta hai bina manually refresh kiye. Bahut time bachta hai.

Frontend Ki Files

File	Kahan Hogi	Kya Kaam Karti Hai
App.jsx	frontend/src/	Main component — sab kuch yahan se start hota hai
ChatBox.jsx	frontend/src/components/	Main chat window — messages, input, suggestions
Message.jsx	frontend/src/components/	Ek individual message bubble — user ya AI
Header.jsx	frontend/src/components/	Top bar — ArjunAI naam, live indicator

File	Kahan Hogi	Kya Kaam Karti Hai
Suggestions.jsx	frontend/src/components/	Quick question buttons — click karo, poochho
useArjunAI.js	frontend/src/hooks/	API call logic — backend se baat karna
index.css	frontend/src/	Tailwind CSS import karo yahan

3. BACKEND KYA HAI?

Jo Andar Kaam Karta Hai

Backend Ki Definition

Backend woh cheez hai jo user ko nahi dikhti lekin andar ka saara zaruri kaam karta hai. Jab user React chatbot mein sawaal likhta hai aur Send dabata hai — yeh sawaal internet ke zariye server pe jaata hai. Server pe backend chal raha hota hai. Backend woh sawaal leta hai, AI se jawab maangta hai, aur wapas bhejta hai. Backend server pe chalta hai — user ke computer pe nahi.

FastAPI — Kya Hai Aur Kyun Use Karte Hain?

FastAPI ek Python web framework hai. Framework matlab ek ready-made structure jo common kaam already kar deta hai — aapko sirf apna specific logic likhna hota hai. FastAPI teen wajah se choose kiya. Pehli wajah — speed. FastAPI Python ka fastest web framework hai — ek saath hazaron requests handle kar sakta hai. Doosri wajah — simplicity. Code likhna bahut simple hai — Python ki type hints se automatically validation hoti hai. Teesri wajah — AI libraries. Sab major AI libraries Python mein hain — Gemini, Claude, OpenAI sab ka Python SDK hai. Backend bhi Python mein rakhne se sab ek jagah rahega.

Python — Kya Hai Aur Kyun?

Python ek programming language hai. Yeh duniya ki sabse popular language hai AI aur data science ke liye. Iska syntax bahut simple hai — English jaisa padhta hai. Sab AI companies — Google, Anthropic, OpenAI, xAI, Groq — apna official Python SDK deti hain. Matlab ArjunAI ka backend Python mein hoga aur charon AI agents ke saath baat karna easy hoga.

Redis — Kya Hai Aur Kyun Zaroori Hai?

Redis ek in-memory data store hai. Memory mein matlab RAM mein data rakhta hai — isliye bahut fast hai. ArjunAI mein Redis caching ke liye use hoga. Caching ka matlab hai jawab ko save karna taaki dobara wahi sawaal aaye toh AI ko call na karni pade. Practical example: agar 500 users ek din mein 'Nifty 50 ka P/E kya hai' poochhein — bina Redis ke 500 baar Gemini call hogi. Redis ke saath pehli call ka jawab 2 ghante ke liye save hoga — baaki 499 users ko instantly wahi jawab milega. Paise bachte hain, speed badhti hai.

Analogy

Socho ek library ka librarian hai. Pehli baar koi book maangta hai — librarian jaa ke dhundh ke laata hai. Doosri baar koi wahi book maangta hai — librarian ne already counter pe rakh di hai — instantly de deta hai. Redis wahi counter hai.

Backend Ki Files

File	Kya Kaam Karta Hai
main.py	FastAPI server — requests receive karo, route karo, respond karo
ai_router.py	4 AI agents ka cascade logic — kaunsa pehle, kaunsa fallback
system_prompt.py	ArjunAI ke rules — sirf stocks, Hindi mein, disclaimer hamesha
cache.py	Redis caching — same query cache karo, paise bachao

File	Kya Kaam Karta Hai
rate_limit.py	Free users: 5 queries/day, Pro users: unlimited
requirements.txt	Sab Python libraries list — pip install se install hongi
.env	API keys — KABHI GitHub pe mat daalna

4. AI AGENTS KYA HAIN?

4 AI Models — Cascade Mein

AI Agent Ka Matlab

AI agent matlab ek AI model ka API. API matlab Application Programming Interface — ek tarika jisse aap kisi service se programmatically baat kar sakte ho. Jaise aap Gemini ke API ko Python se call karte ho — question dete ho, jawab milta hai. ArjunAI mein 4 alag AI agents hain. Woh cascade mein try hote hain — pehle free wala, agar woh fail kare toh doosra, aur phir teesra, aur finally Pro users ke liye Claude.

Cascade Kyun? Waterfall Kyun?

Ek akela AI agent use karna easy hota, lekin risky bhi. Agar Gemini ka server down ho — poora ArjunAI band ho jaata. Cascade mein charon agents hain — ek fail kare toh doosra kaam karta hai, users ko pata bhi nahi chalta. Aur cost bhi optimize hoti hai — sabse sasta pehle try hota hai.

Agent	Provider	Free Limit	Cost After	Speed	Best For	Signup
Gemini 2.0 Flash	Google	1,500 req/day FOREVER	Rs.0 always	2-3 sec	Main AI — best free quality	aistudio.google.com
Grok 4.1 Fast	xAI (Elon)	\$175/month free (\$25 signup + \$150)	~Rs.17/1M tokens	1-2 sec	Fallback + real-time X data	console.x.ai
Groq Llama 3.3	Groq	1,000 req/day FOREVER	Pay per use	0.5-1 sec	Fastest — 10x GPU se tez	console.groq.com
Claude Haiku 4.5	Anthropic	\$5 trial only	\$1/1M input tokens	2-4 sec	Pro users — best finance quality	console.anthropic.com

Gemini Flash — Detail Mein

Google Gemini 2.0 Flash sabse pehle try hota hai kyunki 1,500 requests per day hamesha ke liye free milti hain — koi credit card nahi chahiye, koi expiry nahi. Quality GPT-4o ke barabar hai — stock market questions ke liye bahut capable hai. API key leni hai aistudio.google.com pe — Google account se login karo, 'Get API Key' click karo, copy karo. Python mein google-generativeai library se connect hota hai.

Grok — Detail Mein

Grok xAI ka model hai — Elon Musk ki company. Signup karne pe \$25 free credit milta hai. Phir Settings mein 'Data Sharing' ON karo — ismein aap agree karte ho ki aapka anonymized data training mein use ho — iske badle \$150 per month extra credit milta hai. Total \$175 per month free. Ek aur unique feature hai Grok ka — real-time X (Twitter) data access. Stock market news Twitter pe pehle aati hai — Grok usse directly access kar sakta hai. Technical side pe — Grok OpenAI jaisa format use karta hai sirf base_url alag hai. Matlab ek hi openai Python library se Grok bhi connect hota hai.

Groq — Detail Mein

Groq alag company hai — Grok se bilkul alag. Groq ne custom hardware chips banaye hain sirf AI inference ke liye — isko LPU yaani Language Processing Unit kehte hain. Normal GPU se 10 guna fast hai. 1,000 requests per day hamesha free. Llama 3.3 70B model use hoga jo Meta ka open source model hai. Response 0.5 se 1 second mein aata hai — users ko lagta hai typing ho rahi hai turant. Yeh bhi OpenAI format use karta hai.

Claude — Detail Mein

Claude Anthropic ka model hai — AI safety pe focused company. Finance reasoning mein Claude sabse accurate hai — galat information dene ke chances sabse kam hain. Haiku version use karenge jo Claude ka sabse affordable version hai. \$1 per million input tokens — ek sawaal pe approximately Rs.0.05 lagta hai. Pro users ke liye dedicate karenge — woh Rs.499 per month de rahe hain toh best quality deserve karte hain.

COST CALCULATION Teeno free agents — Gemini + Grok + Groq — milake 4,500+ free requests per day dete hain. Redis caching se 80% requests cache se serve hoti hain. Practically ArjunAI ka AI cost pehle 3-4 mahine mein Rs.0 rehta hai.

5. SYSTEM PROMPT KYA HAI?

ArjunAI Ke Rules Aur Personality

System Prompt Ka Matlab

System prompt ek set of instructions hai jo AI model ko dete hain conversation shuru hone se pehle — background mein. User ko yeh nahi dikhta. Yeh instructions batate hain: AI kaun hai, kya kar sakta hai, kya nahi kar sakta, kaise jawab dena hai. ArjunAI ke liye system prompt mein likha hoga ki woh ek Indian stock market expert hai, sirf NSE aur BSE stocks ke baare mein baat karega, Hindi ya Hinglish mein jawab dega, aur hamesha disclaimer dega ki 'yeh financial advice nahi hai.'

Analogy

Socho ek naya employee join karta hai. HR unhe ek employee handbook deta hai — company rules, kaise behave karna hai, kya allowed hai, kya nahi. System prompt wahi handbook hai AI ke liye — job start hone se pehle di jaati hai.

ArjunAI System Prompt Mein Kya Hoga?

Rule	Detail
Sirf stock sawaal	NSE/BSE stocks, Nifty, Sensex, mutual funds, IPOs, ETFs — haan. Crypto, real estate, recipe, weather — nahi.
Refuse karna	Non-stock sawaal pe: 'Main sirf Indian stock market ke sawaalon ka jawab de sakta hoon.' Politely refuse.
Language	Hindi ya Hinglish mein jawab do. User ne Hindi mein poocha toh Hindi mein jawab. Mix karna allowed.
Format	Short paragraphs. Bullet points use karo. Numbers bold mein. Max 200-250 words. Easy to read.
Disclaimer	Har jawab ke baad: 'Yeh financial advice nahi hai — apne advisor se confirm karein.'
Honesty	Galat information mat dena. Agar pata nahi toh clearly bolna: 'Mujhe is specific data ka pata nahi.'
Tone	Knowledgeable lekin approachable. Jaise ek jaakar dost jo stocks jaanta ho.

Chat History Kya Hai?

Akela sawaal dene se AI ko context nahi milta. Example: User pehle poochhe 'Reliance ka P/E kya hai?' — AI jawab de. Phir user poochhe 'Uski ROE kya hai?' — agar sirf yeh doosra sawaal AI ko diya jaaye bina pehle ke, AI nahi samjhega ki 'uski' matlab Reliance ki. Chat history matlab last 4-6 messages AI ko dena — taaki woh samjhe ki conversation mein kya ho raha hai. Yeh conversations mein natural flow banata hai.

6. TOOLS AUR SETUP

Kaunsa Software Kahan Se
Download Karo

Zaroori Tools — Ek Ek Explain

Tool	Kya Hai	Kahan Se Lo	Cost
Python 3.11+	Main programming language — backend isi mein banega	python.org → Downloads	Free
Node.js 20+	JavaScript runtime — React ke liye zaruri	nodejs.org → Downloads	Free
VS Code	Code editor — yahan code likhenge	code.visualstudio.com	Free
Git	Version control — code save aur track karne ke liye	git-scm.com	Free
Postman	API testing tool — backend test karne ke liye	postman.com	Free
GitHub account	Code online store karne ke liye — deploy ke liye bhi	github.com	Free

4 AI APIs — Sign Up Kaise Karein?

AI	Website	Steps	Free Milta Hai
Gemini	aistudio.google.com	Google account se login → 'Get API Key' click karo → Copy karo	1,500 req/day forever
Grok	console.x.ai	Sign up karo → API Keys section → Key banao → Settings mein Data Sharing ON karo	\$25 + \$150/mo = \$175 total
Groq	console.groq.com	Sign up karo → API Keys section → Create API Key → Copy karo	1,000 req/day forever
Claude	console.anthropic.com	Sign up karo → API Keys → Create Key (credit card zaruri hai \$5 trial ke liye)	\$5 trial credit

SECURITY WARNING .env file mein saari API keys store hongii. Yeh file KABHI GitHub pe mat daalna. .gitignore file mein .env add karo — isse Git automatically ignore kar dega. API key leak hone pe immediately us key ko delete karke nayi banao.

7. 6 WEEK DEVELOPMENT PLAN

Har Hafte Exactly Kya Karna Hai

Week 1 Environment Setup + Project Structure

Foundation banao — sab tools ready karo

Pehle hafte mein koi code nahi likhna hai. Sirf environment setup karna hai — sab tools install karo, API keys lo, basic project structure banao. Yeh ek baar sahi se kar lo toh baaki sab weeks smooth rahenge.

Day 1	Tools Install Karo	Python 3.11+ — python.org se. Node.js 20+ — nodejs.org se. VS Code — code.visualstudio.com se. Git — git-scm.com se. Sab free hain. Install hone ke baad terminal mein check karo: python --version, node --version, git --version — version numbers dikhne chahiye.
Day 2	Project Folder Banao	Ek main folder banao 'arjunai'. Andar do folders — 'backend' aur 'frontend'. Backend folder mein Python virtual environment banao: python -m venv venv. Phir activate karo: source venv/bin/activate (Mac/Linux) ya venv\Scripts\activate (Windows).
Day 3	GitHub Setup Karo	github.com pe account banao. 'arjunai-backend' naam ka repository banao. VS Code mein terminal kholo, backend folder mein jaao, git init karo. .gitignore file banao — usme .env likhdo taaki API keys kabhi upload na hoon.
Day 4	4 AI APIs Sign Up	Gemini: aistudio.google.com — Google account se login, API key lo. Grok: console.x.ai — signup, key banao, Settings mein Data Sharing ON karo (\$175 free). Groq: console.groq.com — signup, key banao. Claude: console.anthropic.com — signup, key banao.
Day 5	FastAPI Basic Setup	pip install fastapi uvicorn python-dotenv. main.py file banao. Ek simple endpoint likho: @app.get('/health') jo 'ArjunAI chal raha hai!' return kare. uvicorn main:app --reload se run karo. Browser mein localhost:8000/health jaao — response aana chahiye.
Day 6-7	.env File + Postman	.env file banao backend folder mein. Charon API keys usme save karo: GEMINI_API_KEY=..., GROK_API_KEY=..., GROQ_API_KEY=..., CLAUDE_API_KEY=... Postman download karo postman.com se. Ek GET request banao localhost:8000/health pe — test karo.

Deliverable: Week 1 end mein: Sab tools installed. 4 API keys ready. FastAPI localhost pe chal raha hai. GitHub pe code hai.

Week 2 AI Router + 4 Agents Connect Karo

Core logic — cascade implement karo

Doosre hafte mein charon AI agents ek ek karke connect honge. Phir unhe ek cascade router mein combine karenge. Yeh ArjunAI ka sabse important part hai — ek fail hone pe doosra automatically try hota hai.

Day 1	Gemini Connect Karo	pip install google-generativeai. ai_router.py file banao. Gemini ko configure karo API key se. Ek simple test karo: Gemini ko 'Reliance Industries ka P/E kya hai?' poochho. Jawab print karo.
Day 2	Grok Connect Karo	pip install openai (Grok OpenAI format use karta hai). OpenAI client banao Grok ke API key se aur base_url='https://api.x.ai/v1'. Model: grok-4-fast. Same test question se test karo.
Day 3	Groq Connect Karo	Same openai library — sirf base_url badlo: 'https://api.groq.com/openai/v1'. Model: llama-3.3-70b-versatile. Test karo — Groq ka response sabse fast aayega.
Day 4	Claude Connect Karo	pip install anthropic. Anthropic client banao Claude API key se. Claude ka format thoda alag — system aur messages alag fields mein hote hain. Test karo same question se.
Day 5	AI Router Class Banao	ArjunAI class banao ai_router.py mein. __init__ mein charon AI initialize karo. ask() method mein cascade logic: pehle Gemini try karo try/except mein. Exception aaye toh Grok try karo. Grok bhi fail kare toh Groq. user_type='pro' ho toh Claude.
Day 6-7	Router Test Karo + Error Handling	Manually GEMINI_API_KEY ko .env se remove karo — Grok try hona chahiye. Grok bhi remove karo — Groq try hona chahiye. Sab restore karo. Every exception properly catch hona chahiye. Meaningful error messages add karo. Postman se test karo.

Deliverable: Week 2 end mein: Charon AI connected. Cascade router kaam karta hai. Postman se question bhejo, jawab aata hai.

Week 3

System Prompt + Cache + Rate Limiting

ArjunAI ko intelligent banao

Teesre hafte mein ArjunAI ko sikhayenge ki woh kaun hai aur kya karna hai. System prompt se uski personality set hogi. Redis cache se paise bachenge. Rate limiting se free aur Pro users ka fark hoga.

Day 1	System Prompt Likho	system_prompt.py file banao. ARJUNAI_PROMPT variable mein poora prompt likho. Rules: sirf stock sawaal, Hindi mein jawab, disclaimer hamesha, galat info pe honest rehna. Prompt ko ai_router.py mein har AI call mein pass karo.
Day 2	Non-Stock Queries Test Karo	'Recipe batao', 'Weather kaisa hai', 'Cricket score kya hai', 'Bitcoin kya hai' — sab test karo. ArjunAI inhe politely refuse karna chahiye. Agar nahi kar raha — system prompt improve karo. Test tab tak karo jab tak sab reject na ho.
Day 3	Redis Setup Karo	pip install redis. upstash.com pe free account banao — serverless Redis, no setup needed. Connection URL milega. Python mein test karo: redis.from_url() se connect karo, ek value set karo, get karo — kaam karna chahiye.
Day 4	Caching Implement Karo	cache.py file banao. Logic: user ka sawaal lo, MD5 hash banao (unique ID), Redis mein check karo is hash pe kuch hai kya. Hai toh wahi return karo. Nahi hai toh AI call karo, jawab Redis mein 7200 seconds (2 ghante) ke liye save karo.
Day 5	Rate Limiting Add Karo	pip install slowapi. main.py mein rate limiter add karo. Free users ke liye: user IP se daily counter track karo Redis mein. 5 se zyada ho jaaye toh 429 error return karo with message 'Pro plan upgrade karein.'
Day 6-7	Chat History Implement Karo	Request body mein 'history' field add karo — list of last messages. AI router mein yeh history messages mein add karo. Test: 'Reliance ka P/E batao' phir 'uski ROE kya hai?' — AI samjhe ki 'uski' matlab Reliance.
Deliverable: Week 3 end mein: ArjunAI sirf stocks ke baare mein baat karta hai. Cache 80% paise bachata hai. Rate limiting kaam karta hai.		

Week 4

React Frontend Banao

Chat UI — jo user dekhega woh banao

Chauthi hafte mein React chatbot UI banayenge. Dark theme, chat bubbles, typing animation, quick suggestion buttons — sab. Is hafte backend se connect nahi karenge — sirf frontend banayenge aur locally test karenge fake data ke saath.

Day 1	React Project Create Karo	npm create vite@latest frontend -- --template react. cd frontend, npm install. npm install tailwindcss postcss autoprefixer. npx tailwindcss init -p. tailwind.config.js configure karo. npm run dev se run karo — localhost:5173 pe dikhna chahiye.
Day 2	Header Component Banao	src/components/Header.jsx banao. ArjunAI naam, green animated dot (live indicator), 'Indian Stock Market Expert' subtitle, Finowings AI branding. Dark background: bg-gray-950. Green text: text-green-400. CSS animate-pulse class for dot.
Day 3	Message Component Banao	src/components/Message.jsx banao. Props: role (user/assistant), content, model. User message: right side, green bubble. AI message: left side, dark grey bubble. 'A' avatar for ArjunAI circle. Model badge — 'Gemini Flash' ya 'Grok 4' small text.
Day 4	ChatBox Main Component	src/components/ChatBox.jsx banao. useState se messages array manage karo. useRef se messages div ka reference lo — scrollIntoView se auto-scroll. Typing indicator: 3 bouncing dots animation jab loading=true ho.
Day 5	Input Area Banao	Text input field — placeholder 'Stock se related sawaal poochho...'. Send button — green color, disabled jab loading ho ya input empty ho. onKeyDown: Enter press hone pe send hona chahiye. Input clear ho jaaye send ke baad.
Day 6-7	Suggestions + Polish	src/components/Suggestions.jsx — 5 quick buttons: 'Reliance analysis', 'Nifty outlook', 'TCS vs Infosys', 'SIP kaise karein', 'Small cap advice'. Click pe automatically send. Mobile responsive check karo — iPhone size pe bhi theek lagna chahiye.

Deliverable: Week 4 end mein: Beautiful dark React UI ready. Typing animation, suggestions, mobile-friendly. Backend se abhi connected nahi.

Week 5

Frontend + Backend Connect + Full Testing

Integration — sab ek saath kaam kare

Paanchve hafte mein frontend aur backend ko connect karenge. Phir poora system thoroughly test karenge — happy path, error cases, edge cases sab. Bugs fix karenge. Is hafte ke end mein ek fully working local chatbot hoga.

Day 1	CORS Setup Karo Backend Mein	Browser ek security rule follow karta hai — ek domain ka frontend doosre domain ke backend se baat nahi kar sakta bina permission ke. localhost:5173 (React) aur localhost:8000 (FastAPI) alag domains hain. FastAPI mein CORSMiddleware add karo — localhost:5173 ko allow karo.
Day 2	API Hook Banao React Mein	src/hooks/useArjunAI.js banao. useState: messages, loading, error. askQuestion function: fetch() se POST request localhost:8000/api/chat pe bhejo. Request body: {question, user_type, history}. Response se answer, model, cached lo.
Day 3	End-to-End Test Karo	Dono servers chalo: uvicorn main:app --reload (backend) aur npm run dev (frontend). Browser mein localhost:5173 jaao. Chat box mein 'Reliance ka P/E kya hai?' type karo. Network tab mein dekho request jaati hai aur response aata hai. Model badge dikhna chahiye — 'Gemini Flash'.
Day 4	Error Handling	Kya hoga agar: network down ho? AI timeout kare? Rate limit cross ho? Backend crash kare? Har scenario ke liye user-friendly error message dikhao. Try-catch everywhere. Console errors ko user ke liye readable message mein convert karo.
Day 5	Different Queries Test Karo	Simple: 'TCS ka P/E ratio kya hai?' Medium: 'Nifty 50 ke top 5 stocks kaun se hain?' Complex: 'Infosys ka fundamental analysis karo.' Non-stock: 'Biryani recipe batao' — yeh reject hona chahiye. Hindi: 'Nifty ka outlook kya hai' — Hindi mein jawab aana chahiye.
Day 6-7	Performance + Final Bugs	Cache test karo: same question 2 baar poochho — doosri baar faster aana chahiye. Model badge sahi show ho raha hai? Rate limit — 5 queries ke baad error aata hai? Mobile pe bhi test karo. Sab bugs note karo aur fix karo. Ek clean pass final polish.

Deliverable: Week 5 end mein: Poora system locally working. Frontend + Backend + 4 AI agents sab connected. Deploy ke liye ready.

Week 6 Deploy + Domain + Beta Launch

Live karo — duniya dekhe ArjunAI ko

Chhathe aur last hafte mein ArjunAI ko internet pe deploy karenge. Free hosting use karenge — Render.com backend ke liye, Vercel frontend ke liye. Domain connect karenge. Beta users ko share karenge aur unka feedback lenge.

Day 1	Backend Deploy — Render.com	render.com pe free account banao. 'New Web Service' click karo. GitHub repo connect karo. Build command: <code>pip install -r requirements.txt</code> . Start command: <code>uvicorn main:app --host 0.0.0.0 --port \$PORT</code> . Environment variables section mein charon API keys add karo. Deploy karo — 5-10 minute lagte hain.
Day 2	Frontend Deploy — Vercel	vercel.com pe free account banao. 'Add New Project' click karo. GitHub repo connect karo. Framework: Vite (automatic detect hota hai). Environment variable add karo: <code>VITE_API_URL</code> = Render.com ka backend URL. Deploy click karo — 2 minute mein live.
Day 3	Domain Connect Karo	namecheap.com ya godaddy.com se .com domain lo — Rs.800/year. Vercel dashboard mein 'Custom Domains' section mein domain add karo. Namecheap mein DNS records update karo — Vercel batayega kaunse records chahiye. HTTPS automatically setup ho jaata hai — Let's Encrypt se free SSL.
Day 4	Production Testing	Live URL pe jaao — domain pe. Browser mein open karo, real queries test karo. Mobile pe bhi test karo. Speed check: pehla response kitni der mein aata hai? Koi issue ho toh Render.com aur Vercel ke logs dekho — error easily dikh jaati hai.
Day 5	Beta Users	5-10 log dhundho jo stocks mein interested hain. WhatsApp pe link bhejo. Unhe poochho: kya jawab accurate lage? Kuch cheez confusing lagi? Koi bug mila? Kaunse features missing hain? Notes lo — yeh sab Phase 2 planning mein kaam aayega.
Day 6-7	Monitoring + V1.0 Declare	uptimerobot.com pe free account banao — 50 monitors free. ArjunAI ke URL ka monitor add karo — har 5 minute mein check hoga. Down hone pe email notification. Feedback notes review karo. Phase 2 (Portfolio + Screener) ka plan shuru socho. ArjunAI Version 1.0 officially live!

Deliverable: Week 6 end mein: ArjunAI live hai internet pe! Real users use kar rahe hain. Monitoring active. Beta feedback aa raha hai.

8. COST + FINAL SUMMARY

Poora Hisaab

Monthly Cost Breakdown

Item	Month 1-3	Month 4-6	Notes
Google Gemini Flash	Rs.0	Rs.0	1,500 req/day hamesha free — forever
Grok 4.1 Fast	Rs.0	Rs.0	\$175/month free credits — Data Sharing ON karo
Groq Llama 3.3	Rs.0	Rs.0	1,000 req/day hamesha free — forever
Claude Haiku 4.5	Rs.0	Rs.0-500	Sirf Pro users ke liye — tabhi lagega
Render.com (Backend)	Rs.0	Rs.0	Free tier available — small projects ke liye
Vercel (Frontend)	Rs.0	Rs.0	Free tier — hobby projects ke liye
Redis (Upstash)	Rs.0	Rs.0	Free 10K req/day — MVP ke liye kafi
Domain (.com)	Rs.67/mo	Rs.67/mo	Rs.800/year — optional lekin professional lagta
UptimeRobot (Monitor)	Rs.0	Rs.0	Free 50 monitors forever
TOTAL	Rs.0-67/mo	Rs.67-567/mo	Almost free — domain optional hai

6 Week Summary

Week	Focus	Kya Milega
Week 1	Setup	Python, Node, VS Code, Git installed. 4 AI API keys ready. FastAPI chal raha hai.
Week 2	AI Router	Charon AI connected. Cascade: Gemini to Grok to Groq to Claude. Postman se tested.
Week 3	Intelligence	System prompt set. Redis cache. Rate limiting. Chat history. Stock-only chatbot.
Week 4	Frontend	Dark React UI. Chat bubbles, typing animation, suggestions. Mobile responsive.
Week 5	Integration	Frontend + Backend connected. Full testing. All bugs fixed. Deploy ready.
Week 6	Launch	Render.com + Vercel deploy. Domain connected. Beta users. Monitoring active.

ARJUNAI — 6 WEEKS MEIN KYA MILEGA:

Ek live, professional, dark-themed AI stock chatbot jo browser mein chal raha hoga. Sirf Indian stock market ke sawaalon ka jawab dega — Hindi ya English mein. 4 AI agents (Gemini + Grok + Groq + Claude) cascade mein — ek fail hone pe doosra automatically try hota hai. Redis cache se paise bachte hain. Rate limiting se free aur Pro users ka fark hoga. Aur sab kuch — Rs.0 se Rs.67 per month mein. Yeh sirf shuruat hai — Phase 2 mein portfolio, screener, aur buy/sell signals aayenge.